SASKEN TECHNOLOGIES

Campus Recruitment – Student Overview & First-Round Question Bank

Role	Software Engineer
Batch	2027 graduates (as per the shared recruitment email)
CTC	■5.0 LPA (as stated in the shared email)
Training	15-hour student preparation programme

Purpose: This guide gives students a clear picture of the company, likely recruitment stages, preparation syllabus and a focused C/C++ question bank for first-round revision.

Important: The recruitment email shared by the placement team is the primary source for the 2027 process details. Sasken has not provided an official question count, test duration, assessment platform or complete test blueprint in that email. Previous candidate experiences vary by campus and year. Therefore, the exam pattern and question bank below are preparation guidance—not an official or guaranteed paper.

Prepared for student preparation and classroom distribution

1. About Sasken Technologies

Sasken Technologies is an Indian product engineering and digital transformation company. Its work spans software and engineering services for technology-led industries. The company describes its work as concept-to-market and chip-to-cognition engineering.

Key business areas

Domain	What students should understand
Semiconductor	Engineering around chips, embedded platforms and related software.
Automotive	In-vehicle infotainment, connected vehicles, body electronics, instrument clusters and ADAS.
Telecom & communication	Cellular networks, 4G/5G-related systems and communication devices.
Industrial & IoT	Connected products, industrial software and device communication.
Consumer / enterprise devices	Software for smart devices and connected products.
Satellite communications	Communication technologies and related engineering solutions.

What the role may involve: Depending on the assigned business unit and project, Software Engineers may work on software development, testing, integration, debugging and product sustenance. The shared email says project allocation will depend on the candidate's skills and aptitude.

Recruitment details from the shared email

Item	Information stated in the email
Position	Software Engineer
Openings	Approximately 200+ engineers for 2027 (company estimate in the email)
Locations	Bengaluru, Kolkata, Pune and Chennai
Initial training	4–8 months; self-study, classroom and on-the-job learning; periodic assessment
Salary	Total ■5.0 lakh per annum
Service agreement	3 years; email states ■5,00,000 payable in case of breach. Read the actual agreement carefully before accepting.
Eligibility	At least 70% in 10th, 12th/Diploma, Bachelor's and Master's if applicable; eligible branches listed in the email; no backlog condition for final-year candidates.

2. Expected Recruitment / Exam Pattern

The following is a practical preparation model based on the stages explicitly mentioned in the shared Sasken email, supplemented by candidate-reported experiences from earlier drives.

Stage	What to expect / prepare
1. Programming assessment	C or C++ programming; implementation of data structures and algorithms; code logic, debugging and output tracing.
2. Technical + aptitude assessment	Core technical topics, abstract reasoning and English communication. Numerical and logical aptitude are useful preparation areas.
3. Technical interview	C/C++ fundamentals, pointers, memory, OOP, DSA, problem-solving, project explanation and possibly OS / computer fundamentals.
4. HR interview	Communication, motivation, location flexibility, training expectations, career goals and understanding of the service agreement.

Possible assessment components reported in previous candidate experiences

- C/C++ output prediction and debugging
- Programming questions based on arrays, strings, searching, linked lists and other DSA topics
- Technical MCQs covering C/C++, OOP and computer fundamentals
- Aptitude, analytical / abstract reasoning and verbal ability
- Some older campus experiences mention a group discussion; the current shared email does not confirm GD.

Do not assume a fixed number of questions or test duration. Students should practise both timed MCQs and timed coding tasks. The actual 2027 pattern may differ by campus and hiring requirement.

3. Expected Syllabus – First Round & Interviews

Area	Topics to prepare
C programming	Data types, operators, loops, functions, arrays, strings, pointers, pointer arithmetic, double pointers, structures, unions, storage classes, preprocessor, recursion, dynamic memory allocation, file handling, bitwise operators, output prediction and debugging.
C++ programming	Classes and objects, access specifiers, constructors / destructors, this pointer, inheritance, polymorphism, function overloading / overriding, virtual and pure virtual functions, abstract classes, pointers vs references, templates, exception handling, STL basics.
DSA	Arrays, strings, linked lists, stacks, queues, searching, sorting, hashing, recursion, trees, BST, traversals, basic graphs, time and space complexity.
Aptitude / reasoning	Number series, percentages, ratios, averages, profit and loss, time and work, speed-distance-time, data interpretation, arrangements, directions, blood relations, patterns and figure-based abstract reasoning.
English	Reading comprehension, grammar, sentence correction, vocabulary, sentence rearrangement and clear spoken communication.
Interview fundamentals	OOP, OS basics (process/thread, scheduling, deadlock, memory), DBMS basics, computer networks (OSI, TCP/UDP, DNS), project explanation and debugging approach.

4. C Programming – 50 Important Questions

Use these as first-round revision prompts. Students should be able to explain each concept and write a small example where applicable.

1. What is the purpose of the main() function?

Revision answer: Program execution entry point in a hosted C program.

2. What is the difference between declaration and definition of a variable?

Revision answer: Declaration introduces its name/type; definition allocates storage (for an object).

3. What is the difference between = and ==?

Revision answer: Assignment versus equality comparison.

4. What is the difference between ++i and i++?

Revision answer: Pre-increment changes then yields; post-increment yields old value then changes.

5. What is the sizeof operator used for?

Revision answer: Returns the size in bytes of an object or type.

6. How is sizeof different from strlen()?

Revision answer: sizeof measures object/storage size; strlen counts characters before the null terminator.

7. What is an array in C?

Revision answer: A contiguous sequence of elements of the same type.

8. What is the relationship between an array name and a pointer?

Revision answer: In most expressions an array converts to a pointer to its first element; they are not identical types.

9. What is a string in C?

Revision answer: A character array terminated by the null character '\0'.

10. What does the null character represent?

Revision answer: The string terminator with value zero.

11. What is a pointer?

Revision answer: A variable that stores the address of an object or function.

12. What does the address-of operator & do?

Revision answer: Produces the address of an object.

13. What does the dereference operator * do?

Revision answer: Accesses the object pointed to by a valid pointer.

14. What is a NULL pointer?

Revision answer: A pointer value guaranteed not to point to an object or function.

15. What is a void pointer?

Revision answer: A generic object pointer that can hold a converted object pointer; it must be converted before dereferencing.

16. What is a dangling pointer?

Revision answer: A pointer referring to an object whose lifetime has ended.

17. What is a wild pointer?

Revision answer: An uninitialized pointer whose value is indeterminate.

18. What is a double pointer?

Revision answer: A pointer that stores the address of another pointer.

19. What is pointer arithmetic?

Revision answer: Arithmetic that advances or retreats by units of the pointed-to type.

20. What is the difference between call by value and using pointer arguments?

Revision answer: C passes arguments by value; a pointer value can be passed to let a function modify the pointed-to object.

21. What is recursion?

Revision answer: A function calling itself directly or indirectly.

22. Why is a base case needed in recursion?

Revision answer: To stop recursive calls and prevent unbounded recursion.

23. What is a function prototype?

Revision answer: A declaration specifying a function's name, return type and parameter types.

24. What is the difference between local and global variables?

Revision answer: Local variables have block scope; global/file-scope identifiers are declared outside functions.

25. What is a static local variable?

Revision answer: A block-scope variable with static storage duration that retains its value between calls.

26. What does extern indicate?

Revision answer: A declaration referring to an object or function defined elsewhere, subject to linkage rules.

27. What is the difference between structure and union?

Revision answer: Structure members have separate storage; union members share storage.

28. Can a structure contain another structure?

Revision answer: Yes; structures can contain members that are structures.

29. What is an enum?

Revision answer: A user-defined enumeration type consisting of named integer constants.

30. What is a typedef?

Revision answer: A mechanism to introduce an alias for a type.

31. What does malloc() do?

Revision answer: Allocates a requested number of bytes; contents are uninitialized.

32. What does calloc() do?

Revision answer: Allocates an array-sized block and initializes all bytes to zero.

33. What does realloc() do?

Revision answer: Resizes a previously allocated block, possibly moving it.

34. What does free() do?

Revision answer: Releases dynamically allocated storage.

35. What is a memory leak?

Revision answer: Allocated memory that is no longer reachable or released when no longer needed.

36. What is undefined behaviour?

Revision answer: Behaviour for which the C standard imposes no requirements.

37. What is a segmentation fault commonly associated with?

Revision answer: Invalid memory access; it is a common operating-system symptom, not a C-defined outcome.

38. What is the difference between break and continue?

Revision answer: break exits the nearest loop/switch; continue skips to the next loop iteration.

39. What is the switch statement used for?

Revision answer: Multiway selection based on an integral or enumeration expression.

40. What is the purpose of the const qualifier?

Revision answer: Prevents modification through that particular lvalue expression.

41. What is a macro?

Revision answer: A preprocessor replacement rule introduced using #define.

42. What is the difference between #include <...> and #include "..."?

Revision answer: They use different header-search rules, which are implementation-defined.

43. What is the purpose of a header file?

Revision answer: Shares declarations, macros, type definitions and interfaces across source files.

44. What is the bitwise AND operator?

Revision answer: & performs AND on corresponding bits of integer operands.

45. What does the XOR operator do?

Revision answer: ^ produces 1 for bit positions where the operands differ.

46. What does the left-shift operator do?

Revision answer: Shifts the left operand's bits left, subject to integer and language rules.

47. What is the difference between signed and unsigned integers?

Revision answer: They represent signed versus non-negative ranges; unsigned arithmetic wraps modulo one more than its maximum.

48. What is a file pointer?

Revision answer: A FILE* handle used by the C standard I/O library.

49. What is the purpose of fopen() and fclose()?

Revision answer: Open a stream and close it.

50. What is the difference between fgetc() and fgets()?

Revision answer: fgetc reads one character; fgets reads a line or up to a specified limit.

5. C++ Programming – 50 Important Questions

Use these as first-round revision prompts. Students should be able to explain each concept and write a small example where applicable.

1. What is C++?

Revision answer: A general-purpose, multi-paradigm programming language that supports procedural and object-oriented programming.

2. What is a class?

Revision answer: A user-defined type that groups data and functions.

3. What is an object?

Revision answer: An instance of a class.

4. What are access specifiers in C++?

Revision answer: public, private and protected.

5. What is encapsulation?

Revision answer: Bundling data and methods while controlling access to internal state.

6. What is abstraction?

Revision answer: Exposing essential behaviour while hiding implementation details.

7. What is inheritance?

Revision answer: Creating a derived class from an existing base class.

8. Name common forms of inheritance.

Revision answer: Single, multilevel, multiple, hierarchical and hybrid.

9. What is polymorphism?

Revision answer: One interface or operation taking different forms.

10. What is compile-time polymorphism?

Revision answer: Polymorphism resolved during compilation, such as overloading.

11. What is runtime polymorphism?

Revision answer: Dynamic dispatch, commonly through virtual functions.

12. What is function overloading?

Revision answer: Same function name with different parameter lists in the same scope.

13. What is function overriding?

Revision answer: A derived class provides a matching virtual member-function implementation.

14. What is a constructor?

Revision answer: A special member function used to initialize an object.

15. Can constructors be overloaded?

Revision answer: Yes, with different parameter lists.

16. What is a copy constructor?

Revision answer: A constructor that initializes an object from another object of the same class.

17. What is a destructor?

Revision answer: A special member function called when an object's lifetime ends.

18. Can a destructor be overloaded?

Revision answer: No; a class has one destructor, though it can be virtual.

19. What is a virtual function?

Revision answer: A member function supporting dynamic dispatch through a base-class interface.

20. What is a pure virtual function?

Revision answer: A virtual function declared with = 0; it makes the class abstract unless overridden as required.

21. What is an abstract class?

Revision answer: A class that cannot be instantiated, typically because it has at least one pure virtual function.

22. Why use a virtual destructor?

Revision answer: To ensure derived destructors run when deleting an object through a base-class pointer.

23. What is the this pointer?

Revision answer: A pointer available in non-static member functions referring to the current object.

24. What is the difference between class and struct in C++?

Revision answer: Default member access and inheritance are private for class and public for struct.

25. What is the difference between pointer and reference?

Revision answer: A pointer stores an address and can be reseated; a reference is an alias and must be initialized.

26. What is the difference between new and malloc()?

Revision answer: new allocates and initializes/constructs typed objects; malloc allocates raw bytes and does not call constructors.

27. What is the difference between delete and free()?

Revision answer: delete invokes destructors and releases new-allocated storage; free releases malloc-family storage.

28. What is operator overloading?

Revision answer: Giving an existing operator a user-defined meaning for class or enumeration types.

29. What is multiple inheritance?

Revision answer: A class derives from more than one base class.

30. What is the diamond problem?

Revision answer: Ambiguity or duplicated base subobjects in a diamond-shaped inheritance hierarchy; virtual inheritance can address shared-base duplication.

31. What is a friend function?

Revision answer: A non-member function granted access to a class's private and protected members.

32. What is a static data member?

Revision answer: A class member shared by all objects of that class.

33. Can a static member function access non-static members directly?

Revision answer: No; it has no this pointer and needs an object to access non-static members.

34. What is a namespace?

Revision answer: A named scope used to organize identifiers and reduce naming conflicts.

35. What is a template?

Revision answer: A mechanism for writing generic functions or classes.

36. What is function template specialization?

Revision answer: A tailored implementation for a particular template argument pattern.

37. What is exception handling in C++?

Revision answer: Handling exceptional conditions using try, throw and catch.

38. What is the difference between throw and catch?

Revision answer: throw signals an exception; catch handles a matching exception.

39. What is RAII?

Revision answer: Resource Acquisition Is Initialization: resource lifetime is tied to object lifetime.

40. What is the STL?

Revision answer: The Standard Template Library, including containers, algorithms, iterators and related utilities.

41. Name common STL containers.

Revision answer: vector, list, deque, map, set, unordered_map, unordered_set and queue/stack adaptors.

42. What is the difference between vector and list?

Revision answer: vector stores elements contiguously; list is a doubly linked sequence.

43. What is the difference between map and unordered_map?

Revision answer: map is ordered by key; unordered_map is hash-based and has average constant-time lookup.

44. What is an iterator?

Revision answer: An object that provides a way to traverse elements in a container or range.

45. What is the difference between stack and queue?

Revision answer: Stack is LIFO; queue is FIFO.

46. What is a lambda expression?

Revision answer: An inline anonymous function object.

47. What is const correctness?

Revision answer: Using const to express and enforce non-modifying interfaces where appropriate.

48. What is an lvalue?

Revision answer: An expression that identifies an object or function; in modern C++ value categories are more nuanced.

49. What is a move constructor?

Revision answer: A constructor that can take resources from an rvalue source, commonly via an rvalue reference.

50. What is the purpose of std::unique_ptr?

Revision answer: A smart pointer providing exclusive ownership of a dynamically managed object.

6. Student Revision Plan

Use this seven-day plan after the classroom training or in the week before the recruitment assessment.

Day	Revision target	Suggested practice
1	C fundamentals and operators	Revise data types, loops, functions; solve 15 output questions and 3 basic programs.
2	Pointers and memory	Practise pointer arithmetic, arrays/pointers, malloc/calloc/realloc/free and debugging.
3	C++ and OOP	Revise classes, inheritance, polymorphism, constructors, virtual functions and STL.
4	Arrays, strings and searching	Solve 5 timed coding questions; revise complexity.
5	Linked lists, stacks, queues and trees	Solve 4 implementation/traversal problems.
6	Aptitude and technical fundamentals	Practise abstract reasoning, arithmetic, English, OS and networking MCQs.
7	Mock assessment and interview	Attempt a timed mock; review mistakes; practise self-introduction and project explanation.

Daily 3-hour routine: 60 minutes C/C++ concepts and output questions + 60 minutes DSA coding + 30 minutes aptitude/reasoning + 30 minutes technical MCQs or interview revision.

7. Interview Preparation Checklist

Students should be ready to answer and explain:

- Tell me about yourself (60–90 seconds).
- Why do you want to join Sasken?
- Explain one academic/personal project and your individual contribution.
- Explain a technical challenge you faced and how you debugged it.
- Explain pointers, dynamic memory, OOP and one data structure with examples.
- Solve a simple coding problem while explaining the approach and complexity.
- Are you comfortable with the stated work locations and initial training period?
- Have you read and understood the service-agreement terms?

8. Exam-Day Strategy

1. Read each question carefully and identify what is being asked.
2. In coding, first confirm input/output format and constraints.
3. Test edge cases: empty input, one element, duplicates, negative values and large values where relevant.
4. Prefer a correct, clear solution before attempting unnecessary optimization.
5. For MCQs, avoid spending too long on one difficult question; follow the actual test's navigation and marking rules.
6. In interviews, explain your reasoning honestly. If unsure, state what you know and how you would investigate.

9. Reference & Accuracy Note

Primary source for role, eligibility, salary, locations, training duration, selection stages and service agreement: the Sasken recruitment email shared by the placement team. Company overview: Sasken official website – https://www.sasken.com/ .

Historical interview experiences can be found by searching candidate-reported Sasken campus interview experiences on GeeksforGeeks, Glassdoor and TakeUForward. Such accounts are anecdotal and may relate to different roles, campuses or years. They should be used only to identify useful practice topics, not as confirmation of the 2027 test

format.

Final reminder: This is an overview and revision guide, not an official Sasken paper. The placement team should share any updated official instructions or test blueprint when received.